

Research on QC-LDPC Decoding Method With Low Quantization Word Length Based on Adaptive Information Mapping in Passive Optical Network

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Abstract—Passive Optical Networks (PON) have advantages such as stable performance, convenient installation, high bandwidth, and resource saving, making them a mainstream network access technology with broad application prospects. The development of such as 8K video, digital twins, VR and other technologies causes explosive growth of network traffic and drives the next generation of PON to evolve towards higher rates, which results in the use of forward error correction (FEC) coding with higher coding gain to improve the power budget of PON. Quasi-cyclic low density parity check (QC-LDPC) codes are widely utilized in PON systems due to high coding gain and parallel encoding and decoding capabilities. However, the application of turbo-decode message passing (TDMP) decoding method is inevitably equated with high hardware complexity of the decoder caused by storing and processing massive information, which is one of the obstacles to PON evolution. Reducing the quantization word length is effective to reduce hardware complexity, but it also leads to saturation of decoding information, causing errors and affecting decoding performance. This work proposes a nonlinear mapping method which utilizes adaptive compression of decoding information through node saturation state monitoring during the decoding process, in order to ensure that the probability density of node information has a reasonable distribution. The simulation results indicate that the method can effectively mitigate the decline in decoding performance under low-word-length conditions.

Index Terms—Quasi-cyclic low density parity check codes, turbo-decode message passing, quantization, nonlinear mapping.

I. INTRODUCTION

IN RECENT decades, with the development of society and advancements in technology, there has been a rapid growth in the demand for bandwidth. According to

Moore's Law, the bandwidth demand of users increases by an order of magnitude every three years. In the field of access networks, the 10G EPON technology has matured and entered commercial deployment as a replacement for 1G EPON. Furthermore, as early as 2011, the ITU-T initiated research on NG-PON2 technology, indicating that PON standards have already entered the era of 40 Gbps [1]. In the face of the ever-increasing bandwidth demand, the capacity of PON systems has to be increased, resulting in higher noise power and subsequently a decrease in the SNR. This poses a significant challenge to the power budget of PON systems. FEC technology, as an essential component of PON, can effectively improve the bit error rate by correcting errors in the codewords using redundant information. It is an economical and effective technique to enhance the power budget of the link. In the early days of optical communication systems, FEC was not commonly employed. However, even the use of a simple FEC scheme can yield significant SNR gains, making FEC highly promising. In 1988, a fundamental FEC scheme specifically designed for optical communication was proposed, enhancing the transmission performance of the system [2].

The application of FEC in PON systems has always encountered challenges due to the inherent asymmetry of PON, where the Optical Network Unit (ONU) located at the end-user premises can only support limited complexity in hardware implementation. As a result, early deployed PON systems primarily relied on simplistic FEC schemes, most of which were based on Reed-Solomon (RS) codes proposed by Irving Reed and Gustave Solomon [3]. Currently, more complicated FEC schemes, such as LDPC codes or concatenated code [7], have been proposed in higher performance PON standards. These advanced FEC schemes provide superior error correction capabilities and enable higher transmission efficiency in PON systems.

LDPC codes have experienced remarkable development since their introduction in the 1960s by Robert Gallager [8]. Initially overshadowed by RS codes, LDPC codes gained significant attention in the late 1990s when efficient decoding algorithms were discovered [9]. An enormous amount of results [10], [11] illustrate that the performance of LDPC codes is close to the Shannon limit in AWGN channels. Besides, LDPC codes have low implementation complexity, as well as the ability to perform high-speed parallel encoding and decoding.

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Subsequent research focused on improving LDPC code performance and decoding algorithms. Irregular LDPC codes were introduced, offering better error-correction capabilities compared to regular LDPC codes. The check matrix of QC-LDPC codes exhibits quasi-cyclic characteristics, making them simple to construct and flexible in terms of encoding and decoding architectures [12], [13]. As a result, they are widely used in various engineering applications. The aforementioned super FEC scheme used in high-rate DWDM submarine optical cable transmission systems in 2004 is essentially a form of QC-LDPC codes.

In terms of decoding algorithms, LDPC soft-decision decoding schemes have become a research focus due to their generally superior performance compared to hard-decision decoding schemes. The belief propagation (BP) decoding algorithm is one of the most commonly used soft-decision algorithms. Among its derivatives based on the log-likelihood ratio (LLR), Normalized Min-Sum (NMS) and Offset Min-Sum (OMS) algorithms have been widely adopted due to their ease of hardware implementation [14], [15], [16]. Furthermore, various adaptive modified MS algorithms have been extensively studied in recent years [17].

In terms of LDPC decoding scheduling, there are two schemes based on the order of information updates. Two Phase Message Passing (TPMP) scheme involves completely independent sequences for updating check node information and variable node information [18]. On the other hand, in TDMP scheme, check nodes and variable nodes update alternately, resulting in significantly improved convergence speed compared to TPMP [19]. Additionally, TDMP can reduce the amount of stored information, effectively decreasing area and power.

Apart from these studies, quantization schemes also have a significant impact on chip area and power consumption. The letter [20] enhances the reconstruction of coarsely quantized check node messages by utilizing messages from the previous iteration as side-information. Non-uniform quantized decoding algorithm is proposed for LDPC in [21]. Simulation results provide coding gain increases closer to of 0.5 dB in higher SNR and has good error floor at 10^{-5} . The paper [22] proposes and verifies a 5-bit non-uniform quantization (NUQ) LDPC code for LLR processing, and compares it with two other uniform quantization schemes. The work [23] supports the massive machine type communications by synergistically utilizing neural MS decoder to enhance decoding performance, and neural quantizer to reduce computational and memory demands. The bottlenecks of low-precision quantization schemes mainly exist in the following three aspects:

- Narrow quantization range, where the absolute values of probability information gradually increase during the decoding process [24]. Insufficient quantization bit width can cause a large amount of information to be quantified to saturation, thereby weakening the decoding performance.
- Low precision of quantization, leading to insufficient discrimination between the minimum and second minimum values in the MS algorithm, thereby reducing the decoding speed.

- The basic idea of traditional NUQ methods is based on the initial step size of quantization, introducing the concept of growth rate to exponentially increase the quantization interval, thereby increasing the quantization range at the cost of increasing complexity. This method is suitable for decoding message sequences with most values distributed in lower regions and a few values distributed in higher regions. The construction of NUQ algorithms is complex and there are certain difficulties in hardware implementation [25].

This paper addresses the phenomenon caused by low quantization bit width and proposes a nonlinear mapping scheme to effectively compress the decoding information, thereby avoiding the occurrence of quantization saturation in a large number of nodes and low discrimination between the minimum and second minimum values. By continuously monitoring the saturation status of each variable node, the external information transmitted between nodes is compressed to ensure it remains within the quantization range. The simulation results indicate that the method can significantly reduce the word length of the decoding information with small decoding performance loss.

The rest of the paper is organized as follows. In section-II, BP decoding algorithms and related basic knowledge is introduced. Section-III discusses the impact of information quantization schemes on decoding and an nonlinear mapping method for low quantization word length. In section-IV, the effectiveness of the method is demonstrated through simulation. Section V concludes the papers.

II. BP DECODING ALGORITHMS

A. Motivation

In this section, we primarily introduce two forms of the BP decoding algorithm and the decoding scheme used in this article for testing. To enable LDPC codes to be effectively applied in PON systems, it is crucial to select efficient and hardware-friendly decoding schemes. On the other hand, the decoding scheme is also a prerequisite for determining the quantization scheme. The NMS algorithm, as one of the most suitable and efficient LDPC decoding algorithms for hardware implementation, originates from the BP algorithm based probability. The analysis and improvement of subsequent quantization schemes are also based on the Gaussian evolution of the BP algorithm.

B. Abbreviations and Acronyms

The following notations concern message-passing algorithms running on the Tanner graph of an LDPC code, and will be used throughout the paper.

- d_v : the degree of variable nodes;
- d_c : the degree of check nodes;
- l : the number of iterations;
- $-r_{ji}(b)$ ($b = 0, 1$): the external probability information passed from the check node j to the variable node i ; that is, the probability that the check equation j satisfies under the condition that the given variable information and other variable information are independent;
- $q_{ij}(b)$: the external probability information passed from variable node i to check node j , that is, after obtaining

external information of all check nodes and channels except for j , the probability of variable node $c_i = b$ is determined;

- C_i : the set of check nodes connected to variable node i , $C_i = \{j: h_{ji} = 1\}$;
- R_j : the set of variable nodes connected to the check node j , $R_j = \{i: h_{ji} = 1\}$;
- $C_{i \setminus j}$: the set of check nodes connected to variable node i except for j , $C_i = \{j: h_{ji} = 1\}$;
- $R_{j \setminus i}$: the set of variable nodes connected to the check node j except for i , $R_j = \{i: h_{ji} = 1\}$;
- c_{kj} : the k -th bit in the j -th check equation containing c_i ;
- y_{kj} : the received value corresponding to c_{kj} ;
- $P_i(1) = \Pr(c_i = 1 | y_i)$: after receiving y_i , determine the posterior probability that the sending bit (or variable node) is $c_i = 1$;
- S_i : the bits in $\hat{c}$ satisfy the d_c check equations containing c_i ;
- L : the information in the form of LLR;
- K_{ij} : a weighting factor that satisfies a certain relationship in probability.

C. BP Algorithm Based on Probability

Lemma 1: The probability of containing an even number of 1s and containing an odd number of 1s in a binary sequence $\mathbf{a} = (a_1, \dots, a_m)$, where $P(a_k = 1) = p_k$, is:

$$\Pr(\text{Even1s}) = \frac{1}{2} + \frac{1}{2} \prod_{i=1}^m (1 - 2p_k^i) \quad (1)$$

$$\Pr(\text{Odd1s}) = \frac{1}{2} - \frac{1}{2} \prod_{i=1}^m (1 - 2p_k^i) \quad (2)$$

Gallager Theorem:

$$\frac{P(c_i = 0 | \bar{y}, S_i)}{P(c_i = 1 | \bar{y}, S_i)} = \frac{1 - P_i \prod_{j \in C_i} \left(1 + \prod_{i' \in R_j \setminus i} (1 - 2P_{i'}) \right)}{P_i \prod_{j \in C_i} \left(1 - \prod_{i' \in R_j \setminus i} (1 - 2P_{i'}) \right)} \quad (3)$$

From the above formula, it can be concluded that:

$$\frac{P(c_i = 0 | \bar{y}, S_i)}{P(c_i = 1 | \bar{y}, S_i)} = \frac{1 - P_i \prod_{j \in C_i} r_{ji}(0)}{P_i \prod_{j \in C_i} r_{ji}(1)} \quad (4)$$

The external probability information passed from variable node i to check node j can be expressed as:

$$\begin{cases} q_{ij}(0) = (1 - P_i) \prod_{j \in C_i} r_{ji}(0) \\ q_{ij}(1) = P_i \prod_{j \in C_i} r_{ji}(1) \end{cases} \quad (5)$$

Thus, the step of the BP algorithm based on probability is as follows:

S1. Initialization:

Initial probability information:

$$q_{ij}^{(0)}(0) = P_i(0)$$

$$q_{ij}^{(0)}(1) = P_i(1) \quad (6)$$

S2. Hard decision:

$$\hat{c}_i = \begin{cases} 0, q_i^{(0)}(0) > q_i^{(0)}(1) \\ 1, q_i^{(0)}(0) \leq q_i^{(0)}(1) \end{cases} \quad (7)$$

If $H\hat{c} = 0$, output $\hat{c}$ and terminate the decoding;

S3. Check node processing:

$$\begin{cases} r_{ji}^{(l)}(0) = \frac{1}{2} + \frac{1}{2} \prod_{i' \in R_j \setminus i} (1 - 2q_{i'j}^{(l-1)}(1)) \\ r_{ji}^{(l)}(1) = 1 - r_{ji}^{(l)}(0) = \frac{1}{2} - \frac{1}{2} \prod_{i' \in R_j \setminus i} (1 - 2q_{i'j}^{(l-1)}(1)) \end{cases} \quad (8)$$

S4. Variable node processing:

$$\begin{cases} q_{ij}^{(l)}(0) = K_{ij} P_i(0) \prod_{j' \in C_i \setminus j} r_{ji'}^{(l)}(0) \\ q_{ij}^{(l)}(1) = K_{ij} P_i(1) \prod_{j' \in C_i \setminus j} r_{ji'}^{(l)}(1) \end{cases} \quad (9)$$

S5. Stop:

If the maximum number of iteration is reached, terminate the decoding, otherwise, go to step S2.

D. BP Algorithm Based on LLR

The probability message is represented in the form of LLR as follows:

$$L(M) = \ln \frac{M(0)}{M(1)} \quad (10)$$

So, the step of the decoding is:

S1. Initialization:

Initial probability information in the form of LLR:

$$L^{(0)}(q_{ij}) = L(P_i) = \ln \frac{P_i(0)}{P_i(1)} \quad (11)$$

S2. Hard decision:

$$\hat{c}_i = \begin{cases} 0, L^{(L)}(q_i) > 0 \\ 1, L^{(L)}(q_i) < 0 \\ \text{unknown}, L^{(L)}(q_i) = 0 \end{cases} \quad (12)$$

If $H\hat{c} = 0$, output $\hat{c}$ and terminate the decoding;

S3. Check node processing:

$$L^{(l)}(r_{ji}) = 2 \tanh^{-1} \left(\prod_{i' \in R_j \setminus i} \tanh \left(\frac{L^{(l-1)}(q_{i'j})}{2} \right) \right) \quad (13)$$

S4. Variable node processing:

$$L^{(l)}(q_{ij}) = L(P_i) + \sum_{j' \in C_i \setminus j} L^{(l)}(r_{ji'}) \quad (14)$$

S5. Stop:

If the maximum number of iteration is reached, terminate the decoding, otherwise, go to step S2.

E. TDMP-NMS Algorithm

In the traditional BP algorithm, variable node i waits for all the check node in C_i to be updated before updating q_i and q_{ij} . For hardware, updating all check node information in C_i simultaneously is undoubtedly complex. To avoid the delay caused by waiting for all check node updates and to accelerate convergence by quickly utilizing the updated variable node information, the TDMP strategy can be adopted. In the TDMP strategy, the check node information in C_i is updated one by one. Once variable node i receives a specific r_{ji} from C_i , it immediately updates the information using the following equation:

$$\begin{aligned} L^{(l)}(q_{ij}) &= L^{(l-1)}(q_i) - L^{(l-1)}(r_{ji}) \\ L^{(l)}(q_i) &= L^{(l)}(q_{ij}) + L^{(l)}(r_{ji}) \end{aligned} \quad (15)$$

In addition, for check node j , updating the information in C_i requires a complex nonlinear operation as described in the previous section. By using the NMS algorithm, this operation can be simplified to a hardware-friendly linear operation, significantly reducing the implementation cost. The check node update equation for the NMS algorithm is as follows:

$$\begin{aligned} L^{(l)}(r_{ji}) &= \prod_{i' \in R_j \setminus i} \text{sign}(L^{(l-1)}(q_{i'j})) \cdot \alpha \cdot \min_{i' \in R_j \setminus i} (|L^{(l-1)}(q_{i'j})|) \end{aligned} \quad (16)$$

where α is normalized factor.

All experiments in the following text were conducted based on the TDMP-NMS algorithm.

III. GAUSSIAN APPROXIMATION AND INFORMATION QUANTIZATION SCHEMES

A. Motivation

In part B, we introduce Gaussian approximation, in order to discuss the fundamental reasons for the impact of quantization word length on decoding performance in part C. In part D, we compare the differences between two compression schemes and recommend using LUT tables for information compression. Finally, an adaptive information compression decoding scheme is proposed and its pseudocode is provided in part E.

B. Gaussian Approximation

For LDPC codes using LLR-BP decoding algorithm on memoryless binary-input AWGN channels, Chung et al. [24] proposed Gaussian approximation to simplify the density evolution theory of multidimensional check node and variable node messages into one-dimensional mean evolution theory. This approximation facilitates the analysis of the asymptotic performance of LDPC codes. On the other hand, the Gaussian approximation also reveals the fundamental reasons for the impact of quantization schemes on decoding performance. First, we provide a definition:

Definition 1: Let $f(x)$ represent the probability density of message in the form of LLR. The symmetry condition can

be expressed as $f(x) = f(-x)e^x$. If $f(x) \sim N(\mu, \sigma^2)$, then the symmetry condition can be simplified to $\sigma^2 = 2\mu$. In this case, the Gaussian distribution is called the symmetric distribution.

Definition 2: Let $f(x)$ represent the probability density of message in the form of LLR. The symmetry condition can be expressed as $f(x) = f(-x)e^{-x}$. If $f(x) \sim N(\mu, \sigma^2)$, then the symmetry condition can be simplified to $\sigma^2 = -2\mu$. In this case, the Gaussian distribution is called the anti-symmetric distribution.

Definition 3: For a generalized symmetric distribution that includes both symmetric and anti-symmetric distributions, if $\mu = -\mu$, the distribution is called the isomorphic generalized symmetric distribution with respect to μ .

For the AWGN channel, it can be derived that the initial probability information in LLR form is given by:

$$L(c_i | y) = \ln \frac{\Pr(c_i = +1 | y)}{\Pr(c_i = -1 | y)} = \frac{2}{\sigma^2} y \quad (17)$$

Based on the symmetry condition and the central limit theorem, it can be concluded that all types of decoding information follow the isomorphic generalized symmetric distribution with respect to $2/\sigma^2$.

Assuming the degree distribution of irregular LDPC codes are $\lambda(x) = \sum \lambda_i x^i$, $\rho(x) = \sum \rho_i x^i$, the process of Gaussian approximation is as follows:

S1. Initialization:

Initial mean evolution of variable message:

$$\mu_0 = \mu_{v,i}^{(0)} = \mu_v^{(0)} = \frac{2}{\sigma^2} \quad (18)$$

S2. Mean evolution of check message

$$\mu_c^{(l)} = \sum_{i=2}^{d_c} \rho_i \varphi^{-1} \left(1 - \left[1 - \sum_{i=2}^{d_v} \varphi(\mu_{v,i}^{(l-1)}) \right]^{i-1} \right) \quad (19)$$

where $\varphi(x)$ is a function for calculating the mean of composite function variables, which is usually approximated as:

$$\varphi(x) = \begin{cases} \exp(-0.4527x^{0.86} + 0.0218) & x < 10 \\ \sqrt{\frac{\pi}{x}} \left(1 - \frac{10}{7x}\right) \exp\left(-\frac{x}{4}\right) & x \geq 10 \end{cases} \quad (20)$$

S3. Mean evolution of variable message

$$\mu_v^{(l)} = \sum_{i=2}^{d_v} \lambda_i [\mu_0 + (i-1)\mu_c^{(l)}] \quad (21)$$

S4. Calculation the probability of errors

The PDF of the variable message in the l -th iteration is $P(l) v \sim N(\mu(l) v, 2\mu(l) v)$, the error probability is:

$$P_e^{(l)} = 2 \int_{-\infty}^0 P_e^{(l)} dv \quad (22)$$

According to the Gaussian approximation, it is known that during iterations, the mean of the isomorphic generalized symmetric distribution increases continuously. In other words, the probability density bimodality of the variable node messages gradually spreads towards positive and negative infinity. In this study, we focus on LDPC(2304, 1920) codes [26] with a code rate of 5/6 and degrees distribution of

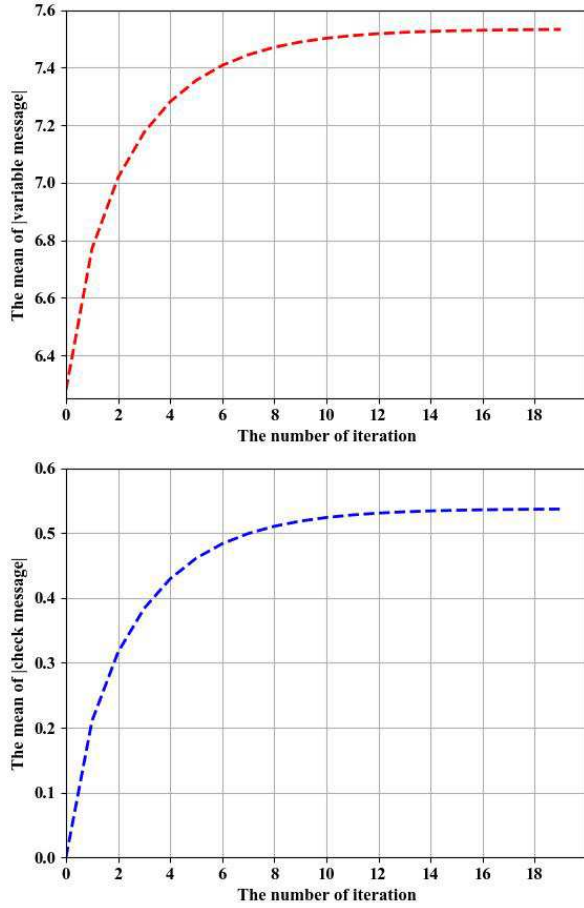


Fig. 1. The trend of the mean of variable and check messages with respect to the number of iterations.

$\lambda(x) = 1/8x^2 + 5/12x^3 + 11/24x^4$ and $\rho(x) = x^{20}$. Fig. 1 shows the variation of the mean of variable node and check node messages with the number of iterations at 4dB.

C. The Impact of Information Quantization Schemes on Decoding

In the case of using floating-point algorithms, LDPC codes exhibit excellent error correction performance. However, practical applications demand the use of quantization with minimal word length to meet performance requirements while satisfying constraints on system power consumption and area. A smaller quantization word length implies a limited quantization range for information without sacrificing quantization precision. When the quantization range of information is limited, the mean of the isomorphic generalized symmetric distribution saturates at the boundaries of the quantization range, leading to saturation quantization errors. If the quantization range is too small, it can severely impact decoding performance.

Similarly, for the aforementioned LDPC codes, different quantization word length are considered to investigate the variation of mean values for various messages during the decoding process. $Q(qv, qc)$ denotes a quantization bit width of qv bits for variable messages and qc bits for check messages. $Q(F, F)$ represents the floating-point algorithm.

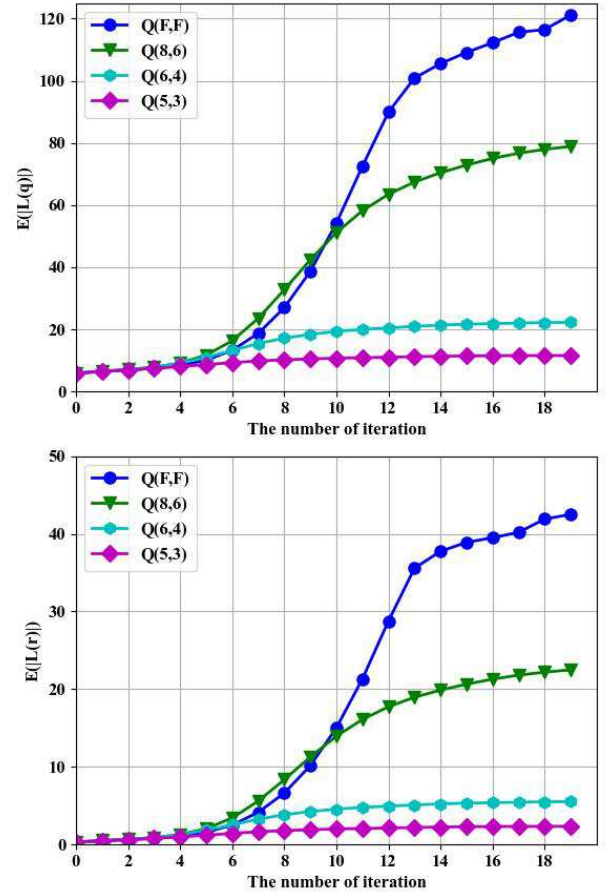


Fig. 2. Variation of the $E(|L(q)|)$ and $E(|L(r)|)$ with respect to the number of iterations under different word length.

Fig. 2 illustrates the variation of mean absolute values for decoding messages of LDPC (2304, 1920) under the conditions of BPSK modulation, AWGN channel, and a SNR of 3dB. It can be observed that the blue curve representing the floating-point algorithm continuously increases during the decoding process, while the remaining curves representing fixed-point algorithms tend to flatten near the boundaries of the quantization range. This indicates the presence of quantization saturation phenomena in the decoded information.

D. Decoding Information Mapping

Regarding the quantization scheme for decoding messages, the analysis above reveals that the low-word-length quantization scheme primarily affects the quantization range of the decoded information. When a majority of decoding messages reach a state of quantization saturation, there will be a lack of discrimination for the larger values of the decoding information, making it extremely challenging to correct the signs of error variable nodes that are in a saturated state. To achieve an expanded quantization range for decoding information under low-word-length condition, the most effective approach is to compress the decoding information.

Assuming the quantization interval of decoding information is I , the quantization word length is w , one sign bit is w_s ,

TABLE I
COMPARISON OF INFORMATION COMPRESSION RESULTS
OBTAINED FROM DIFFERENT SCHEMES

Original information		Directly compress Z		Compression in the numerical domain	
values	Z	values	Z	values	Z
0	0000000	0	0000000	$QM(0)$	0000000
1	0000001	0	0000000	$QM(5/8)=0$	0000000
2	0000010	1	0000001	$QM(10/8)=1$	0000001
3	0000011	1	0000001	$QM(15/8)=2$	0000010
4	0000100	2	0000010	$QM(20/8)=3$	0000011
5	0000101	2	0000010	$QM(25/8)=3$	0000011
6	0000110	3	0000011	$QM(30/8)=4$	0000100
7	0000111	3	0000011	$QM(35/8)=4$	0000100
8	0001000	5	0000101	$QM(40/8)=5$	0000101
...	...	...	...	...	...

and the value of information quantization is Z , then Z can be expressed as a polynomial to the power of 2:

$$Z = I_{w_s} P(Z) = I_{w_s} (z_{w-2} 2^{w-2} + \dots z_1 2^1 + z_0 2^0) \quad (23)$$

where the coefficients in polynomial $P(Z)$ are either 0 or 1, and the binary vector $Z = (z_{w-2}, \dots, z_1, z_0)$ composed of all coefficients and the sign bit w_s can represent a decoding information.

In the TDMP-NMS algorithm, all operations are linear, so simply changing the quantization spacing is equivalent to achieving compression of quantization information. Assuming that the quantization interval is β times larger, and the new quantization information is represented as:

$$Z' = \beta I_{w_s} P(Z') = \beta I_{w_s} (z'_{w-2} 2^{w-2} + \dots z'_1 2^1 + z'_0 2^0) \quad (24)$$

where Z and Z' represent the quantified values of the same information, and the quantization interval cannot be reflected in the decoding process. Therefore, the following relationship is established:

$$Z = I_{w_s} \frac{P(Z)}{\beta} = I_{w_s} P(Z') = Z' \quad (25)$$

the scaling of binary sequence Z is generally achieved in hardware by shifting and summing. Generally speaking, it is necessary to design the following circuits to meet the β of different configurations.

During the simulation process using the aforementioned compression circuit, we observed that certain values of β can affect the accuracy of information compression when the mid-tread quantizer is not used. For example, when $\beta = 8/5$, it is equivalent to Enable_1 and Enable_3 switch "1" in the hardware. Assuming the original quantization interval $I = 1$, $w = 8$, the information obtained after compressing it using the aforementioned hardware circuit and in the numerical domain is presented in Table I.

where $QM(\cdot)$ represents mid-tread quantizer.

Directly shifting and summing Z will result in certain information loss, as shown in the light gray section in Table I. When $\beta = 2^k$, $k \in \mathbb{Z}$, the information compression results of the two schemes are consistent. Therefore, the reason for

the information loss caused by β is attributed to the inability to store quantitative interval information. In addition, when $\beta \neq 2^k/L$, $k, L \in \mathbb{Z}$, the shift summation circuit cannot handle this situation, and the optimal value of β usually depends on the current quantization word length of decoder message and LDPC codebook. To reduce the loss of this type of information, it is possible to directly use a lookup table (LUT) in the numerical domain for mapping from original information to compressed information. Fig. 4 illustrates the circuit implementation of the LUT table, taking 8-bit quantization word length as an example. If the quantization word length decreases, the resources used will be reduced by a power multiple of 2. Usually, corresponding compression values can be calculated according different values of β and configured into the relevant registers through the bus, so as to realize the information compression of any β .

From the above analysis, it can be seen that compressing the decoding information directly through multiplication by a coefficient can be used to avoid extensive quantization saturation of the information. However, an important issue that is still overlooked is that this processing method also leads to a lack of discrimination for small decoding information. The position and magnitude of the smallest and second smallest variable node information in the NMS algorithm are crucial factors affecting the decoding performance and speed. For instance, in the original variable message, the minimum and second minimum values before compression are 3 and 2, respectively. After applying information compression, both values become 1. As a result, the positions of the minimum and second minimum values become ambiguous in the next iteration. It is advisable to record the positions of the minimum values before compression, rather than during the iterative decoding process. Then, when correcting message after compression, two different factors α_{min} and α_{min2} can be used to respectively correct the minimum and second smallest values, which can compensate for the information loss caused by information compression process.

E. Adaptive Information Mapping Decoding (AIM) Scheme Based on TDMP-NMS Algorithms

In practical applications, the timing of information compression cannot be generalized due to variations in SNR and the specific choice of quantization word length. The frequency of information compression should match the rate of growth in the mean absolute value of the decoded messages. When the mean absolute value of the decoded messages increases rapidly, frequent information compression is necessary to prevent quantization saturation. Conversely, when the mean absolute value of the decoded messages increases slowly, infrequent information compression is preferred to avoid information loss caused by compression. Therefore, it is essential to observe the saturation state of each variable node to determine whether information compression should be performed.

Assuming that the quantization saturation state is defined as the information reaching S percent the upper limit of the quantization range, information compression is performed if the proportion of variable nodes reaching the quantization saturation state exceeds η_v of the total number. Optimizing

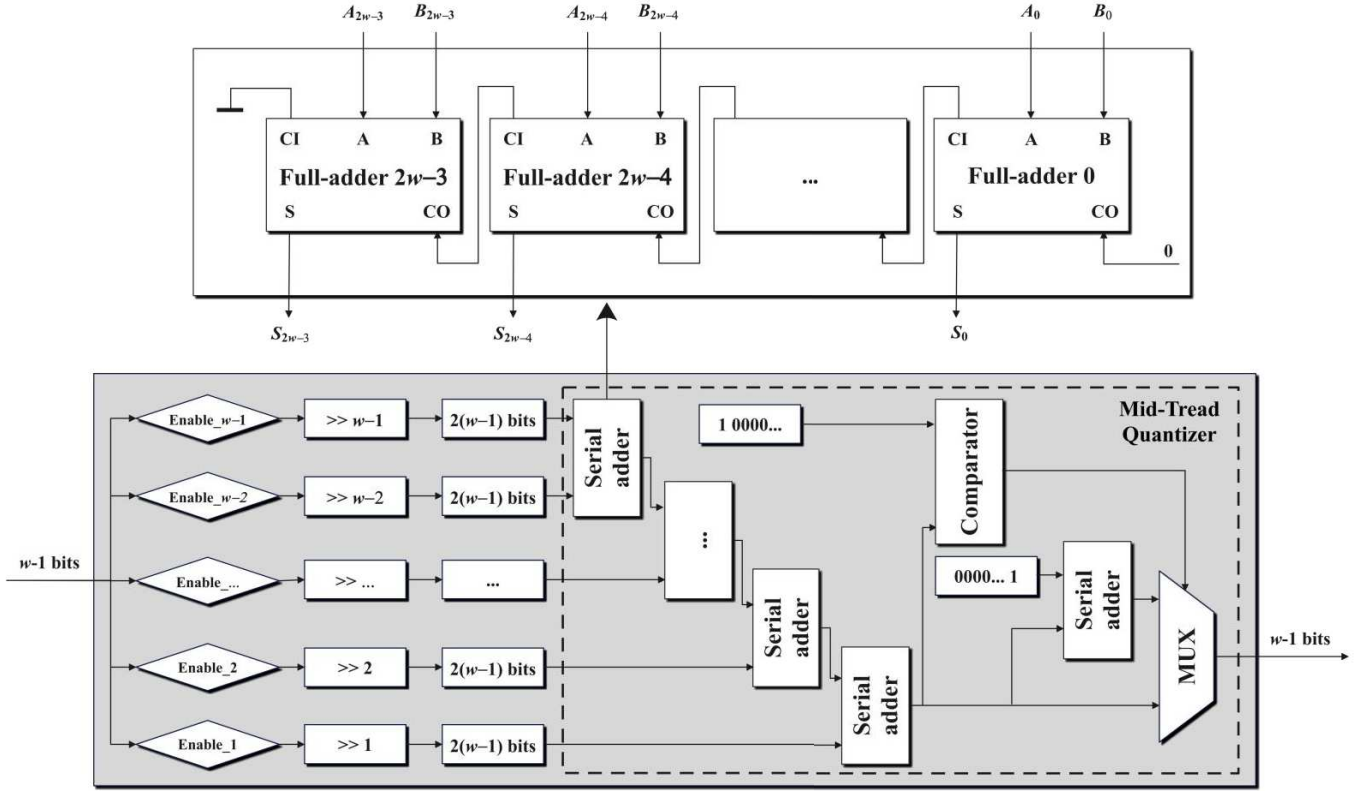


Fig. 3. Schematic diagram of the decoding message compression circuit.

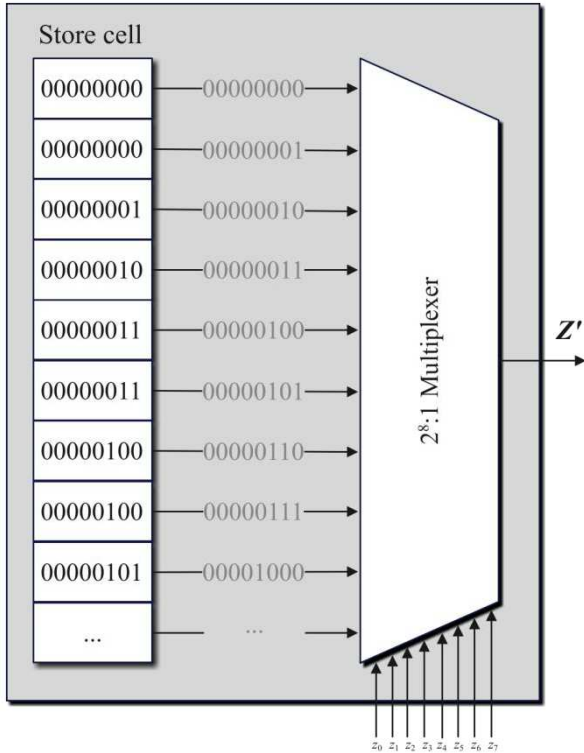


Fig. 4. Schematic diagram of LUT circuit structure.

be calculated by the following equation:

$$S = \frac{2^{w-1} - I}{2^{w-1}} \quad (26)$$

Based on the symmetry condition and the BP decoding algorithm, it is known that the decoding error probability is independent of the transmitted code words. Therefore, for ease of analysis, we can assume that all the transmitted information is +1 after undergoing BPSK modulation. From the analysis in the second section, it is known that the variable messages approximate a Gaussian distribution during the decoding process. Hence, the proportion of variable nodes reaching the quantization saturation state can be expressed using the complementary error function as follows:

$$\eta_{v,i} = \begin{cases} \text{erfc}\left(\frac{2^{w-1} - \mu_{v,i}}{\sqrt{2\mu_{v,i}}}\right), & \mu_{v,i} \leq 2^{w-1} \\ 1 - \text{erfc}\left(\frac{2^{w-1} - \mu_{v,i}}{\sqrt{2\mu_{v,i}}}\right), & \mu_{v,i} > 2^{w-1} \end{cases} \rightarrow \eta_v = \sum_{i=2}^{d_v} \lambda_i \eta_{v,i} \quad (27)$$

The signal energy of the defined variable message is E_v , which can be calculated by the following equation:

$$E_v = \sum_{i=2}^{d_v} \frac{\lambda_i}{4\pi\mu_{v,i}} \int_0^{+\infty} L(q)^2 \exp\left(-\frac{(L(q) - \mu_{v,i})^2}{4\mu_{v,i}}\right) dL(q) \quad (28)$$

the values of S and η_v allows the algorithm to adapt to different SNR and quantization word length. S can generally

After quantization, the portion of the variable message that exceeds the quantization range is defined as quantization

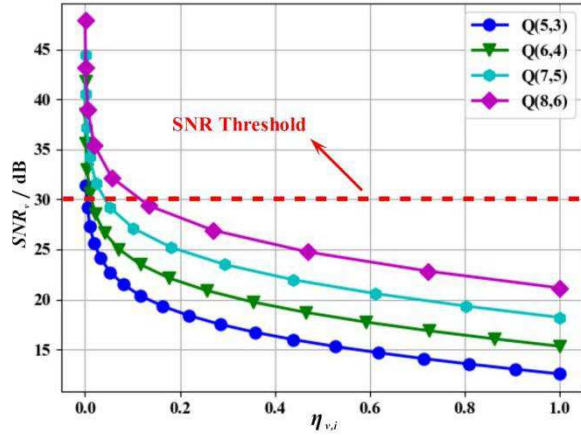


Fig. 5. The relationship between η_v and SNR_v under different quantization word length of variable messages.

saturation noise, with an energy of E_{σ} , which is calculated as:

$$E_{\sigma} = \sum_{i=2}^{d_v} \frac{\lambda_i}{4\pi\mu_{v,i}} \int_{2^{w-1}}^{+\infty} (L(q) - S)^2 \times \exp\left(-\frac{(L(q) - \mu_{v,i})^2}{4\mu_{v,i}}\right) dL(q) \quad (29)$$

Therefore, the SNR_v of variable messages to quantization saturation noise is:

$$SNR_v = 10 \log\left(\frac{E_v}{E_{\sigma}}\right) \quad (30)$$

By taking the quantization word length of the variable messages as 8, 7, 6, and 5 bit, a set of relationships between η_v and SNR_v can be obtained, as shown in Fig. 5. The analysis of figure reveals that for the same η_v , curves with smaller quantization word length exhibit lower saturation SNR_v . When η_v is small, the SNR_v is high, indicating that only a small portion of variable nodes are in the quantization saturation state, resulting in minimal quantization saturation noise. If compression is applied to the decoding messages at this stage, the information that is not in the quantization saturation state will be lost. When $\eta_v \approx 0.1$, the SNR_v for each quantization word length has already dropped below the SNR threshold (SNR degradation slows down when it falls below 30dB, so 30 dB is chosen as the SNR threshold). At this point, if the compression information is not utilized, the decrease in SNR_v will adversely affect the decoding performance.

According to the above process, this work proposed an adaptive information mapping decoding scheme based on TDMP-NMS algorithm, and the decoding process pseudocode is shown as follows.

where Itr_max is the maximum number of iteration, $code_row_num$ represents the number of rows in the check matrix, $code_len$ stands for the length of a code word, and LOC_min is the position of the minimum value.

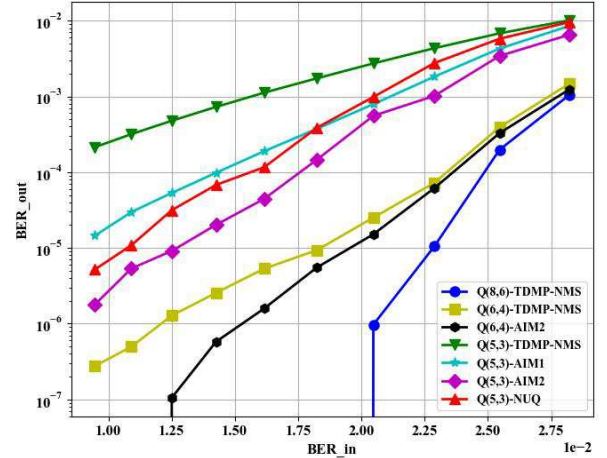


Fig. 6. Comparison of decoding performance between AIM scheme and other scheme.

IV. SIMULATION RESULTS AND ANALYSIS

The maximum number of iteration Itr_max is set to 20, and the saturation proportion threshold η_v is set to 0.1. For the LDPC (2304,1920) code mentioned in Section III, the TDMP-NMS algorithm, the AIM scheme and the NUQ scheme proposed in [21] are employed for decoding under low-word-length condition. AIM1 represents the AIM algorithm without distinguishing the position of the minimum value, while AIM2 represents the AIM algorithm that records the minimum value before information compression and utilizes two different factors correction. Fig. 6 compares the decoding performance of each method.

As the quantization word length decreases from Q(8,6) to Q(5,3), the decoding performance gradually deteriorates, which is due to the increasing severity of quantization saturation phenomena generated by low-word-length schemes during the decoding process. When the AIM2 algorithm is used, the output bit error rate (BER) of Q(5,3) exhibits a noticeable improvement at low input BER levels, while the improvement is not significant at high input BER levels. The two factors correction in the AIM2 algorithm compensates for the information loss caused by information compression, resulting in further enhancement of the decoding performance. Compared to the TDMP-NMS algorithm, the decoding performance can be improved by approximately 1.5 to 2 orders of magnitude at low input BER levels with the Q(5,3) quantization scheme by using AIM2 algorithm. In addition, it can be observed that the AIM2 method performs about 0.5 order of magnitude better than traditional NUQ schemes.

For check node j , the minimum value, second minimum value, position of the minimum value, and symbol information in the set $\{u_{ji}|j \in C_i\}$ need to be stored. For the LDPC (2304,1920) code with $d_c = 20$, 5 bits and 20 bits are required to record the position and symbol information, respectively. In addition, the resources used by the information compression circuit should also be considered in the AIM scheme. The basic circuit units mentioned in the work are compared using standard gate resources, while storage resources are

TABLE II
COMPARISON OF HARDWARE COMPLEXITY AND PERFORMANCE GAIN UNDER DIFFERENT ALGORITHMS AND WORD LENGTH

Decoding and quantization scheme	Variable nodes store resource	Check nodes store resource	Information compression circuit resources		Total gates	Hardware complexity gain	Decoding performance gain (dB) BER _{out} =1e-4
			Store resource	Gate			
TDMP-NMS +Q(8, 6)	8×2304×6	(5+5+5+20)×384×6	\	\	191652	-40.7%	1.7dB
TDMP-NMS +Q(6, 4)	6×2304×6	(3+3+5+20)×384×6	\	\	154368	-13.6%	1.6dB
AIM2 +Q(6, 4)+LUT	6×2304×6	(3+3+5+20)×384×6	2 ⁵ ×5×6	2:1MUX unit: (16+8+4+2+1)×4 11-bit saturation counter: 11×13	155595	-14.5%	1.6dB
TDMP-NMS +Q(5, 3)	5×2304×6	(2+2+5+20)×384×6	\	\	135936	0	0
AIM1 +Q(5,3)+LUT	5×2304×6	(2+2+5+20)×384×6	2 ⁴ ×4×6	2:1MUX unit: (8+4+2+1)×4 11-bit saturation counter: 11×13	136523	0.43%	0.8dB
AIM2 +Q(5,3)+LUT	5×2304×6	(2+2+5+20)×384×6	2 ⁴ ×4×6	2:1MUX unit: (8+4+2+1)×4 11-bit saturation counter: 11×13	136523	0.43%	1.0dB
AIM2 +Q(5, 3)+Shift-SUM	5×2304×6	(2+2+5+20)×384×6	2×4×4×6	4-bit linear shifter: 12×4×4 Full adder: 4×8×5 8-bit comparator: 1×7 2:1MUX unit: 1×4 11-bit saturation counter: 11×13	136634	0.51%	1.0dB

Algorithm 1 Adaptive Information Mapping Decoding Scheme

```

 $\beta \rightarrow \text{LUT}$ 
 $comp = 0$ 
For  $Itr\_num = 1$  to  $Itr\_max$ 
  For  $row = code\_row\_num$ 
    If  $comp = 1$ 
      Two factors  $\alpha_{min}$  and  $\alpha_{min2}$  + NMS algorithm
       $comp = 0$ 
    Else
      NMS algorithm
    End if
     $Num\_V = 0$ 
    For  $i = 1$  to  $code\_len$ 
      If  $V_i^{(Itr\_num)} > S \cdot 2^{w-1}$ 
         $Num\_V^{(Itr\_num)} = Num\_V^{(Itr\_num)} + 1$ 
      End if
    End for
    If  $Num\_V^{(Itr\_num)} > \eta_v \cdot code\_len$ 
       $comp = 1$ 
      Seek for  $LOC\_min$ 
      For  $i = 1$  to  $code\_len$ 
         $L(q_i) = \text{LUT}(L(q_i))$ 
        For  $j \in C_i$ 
           $L(r_{ji}) = \text{LUT}(L(r_{ji}))$ 
        End for
      End for
    End if
  End for
End for

```

converted based on 1 bit = 6 gates. For example, a 2:1 MUX unit uses four gates, a full adder uses five gates, an m -bit comparator uses $7m$ gates, a w -bit linear shifter with a fixed shift number uses approximately $2w$ D flip flops, totaling $12w$ gates, an n -bit saturation counter uses n D flip flops and an n -bit comparator, totaling $13n$ gates. Therefore, Table II provides an estimation of the hardware complexity

and performance gains for various algorithms and quantization word length. The hardware complexity gain is represented by the relative change in the total gates. It should be emphasized that the benchmark test for hardware complexity gain and decoding performance gain is the TDMP-NMS algorithm with Q(5,3). When output BER $\approx$ 1e-4, using the AIM2 algorithm of Q(5, 3), the performance gain is approximately 0.8dB, with a hardware complexity gain of approximately 0.43%. Besides, it is observed that under the condition of Q(6,4), the deployment of AIM algorithm did not significantly improve performance gain, which is different from Q(5,3). One possible reason is that Q(6,4) has a larger quantization range, and the proportion of variable messages reaching saturation is relatively small, which did not trigger information compression. Therefore, in design, whether the benchmark quantization bit width matches the expected hardware complexity gain should be taken in consideration.

V. CONCLUSION

The quantization scheme greatly impacts the performance and hardware complexity of LDPC. Through the Gaussian approximation theory analysis of irregular LDPC codes, it is observed that the mean of absolute value of the decoding message tends to increase with iterations. The primary cause for the deficiency in decoding performance under low quantization word length is the saturation of variable information due to insufficient quantization range.

This paper proposes an effective method to address the inadequate decoding performance caused by low-word-length quantization. By compressing the information, the occurrence of quantization saturation can be avoided. Based on this, we further investigate the hardware implementation scheme for information compression and suggest using a LUT to directly map decoding messages, thereby preventing information loss due to varying compression factors. Furthermore, to effectively counteract the information loss during the compression process, we introduce a two factors correction scheme to compensate for the information compression loss. Finally,

an adaptive approach is proposed that decides whether to compress based on monitoring the saturation state of the variable nodes for different quantization configurations. Simulation results indicate that this scheme can effectively mitigate the decline in decoding performance under low-word-length condition.

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